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ORIGINAL CONTRIBUTION

Integrating Commodity-Based and Trip-Based Approaches of Freight Demand Modelling Using Trip Length Distributions

Aitichya Chandra[1] · Agnivesh Pani[2] · Prasanta K. Sahu[3] · Sushant Sharma[4]

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Abstract The freight demand forecasting process is broadly categorized into trip-based modelling and commodity-based modelling due to the multidimensional units of analysis (i.e., trips, tonnage, volume, etc.). Freight activities can be represented in an improved manner when at least two of these measures can be jointly modelled. The joint modelling concept requires an approximate function to convert tonnage trip length distribution (TLD) to truck (trip) TLD. This research investigates the possibility of such a function for several ISIC commodity classes. Class-specific trip and tonnage lengths are modelled, and, subsequently, approximation functions are developed. The results reveal that approximated trip length distributions match closely with the observed distributions for each class. The approximation functions will be valuable in integrating commodity-based- and trip-based modelling approaches. The inclusion of this function at the trip distribution modelling stage would assist not only in planning for the logistics requirements but also cost-savings in data collection exercises.

✉ Prasanta K. Sahu
prasantsahu222@gmail.com

1 Department of Civil Engineering, Indian Institute of Science (IISc) Bangalore, Bangalore, Karnataka 560012, India

2 Department of Civil Engineering, Indian Institute of Technology (IIT) BHU, Varanasi, Uttar Pradesh 221005, India

3 Department of Civil Engineering, Birla Institute of Technology and Science Pilani, Hyderabad, Telangana 500078, India

4 Texas A&M University, Texas Transportation Institute, Arlington, TX 76013, USA

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Introduction

Trip lengths are crucial for developing a trip length distribution model and are central to the transportation planning process at various levels. It plays a vital role in travel demand estimation, determination of transportation expenditures in an urban region [1, 2], development of freight consolidation programs [3–5], vehicle ownership [6], and design of freight distribution systems [7, 8]. Trip length distribution (TLD) is a measure of the frequency of trips or tonnage (passenger/freight) for a specified interval of several impedance parameters such as distance, time, and generalized cost. TLD function or model describes the variations in passenger or freight demand (trips or tonnage) for specified impedances. Freight TLD functions are essential to all freight demand modeling approaches, such as Class C (Truck model), Class D (four-step commodity model), and Class E (economic activity model) [9], and generally applied at the trip distribution stage. The above models focus either on tonnage (commodity-based model) or truck-trips (trip-based models) as the unit of analysis, limiting the analysis to only one dimension of freight demand [10]. The trip-based model fails to incorporate the underlying economic behavior, which generates the demand (i.e., tonnage) whereas tonnage-based models do not give a realistic account for vehicle activities in an urban area. Advanced freight demand forecasting models such as Class F (Logistic Chain Models) and Class G (Vehicle Touring Models) addressed the limitations of these traditional approaches. These models employ a more

disaggregated approach, unlike the conventional forecasting approaches that use trip distribution modeling to estimate flows between origin–destination (O–D) pairs based on zonal attributes and impedance variables [11]. However, irrespective of the approach, the information on trip or commodity length distribution is necessary for improving the modeling framework and ensuring predictive accuracy. For instance, Class F models have used the gravity model to distribute commodity production to their respective tour types. Class G models (e.g., truck-tour based micro-simulation model) need information on travel distance to generate the number of tours [9]. Thus, accurate estimates of trip lengths remain crucial for freight transportation planning, no matter which modeling approach is used.

Further, commodity-based models focus on the spatial distribution of the amount of freight measured in tons. Data required for such models are obtained from expensive and time-consuming surveys, for instance, the commodity flow survey (CFS) in the United States. On the contrary, truck trip-based models focus on the spatial distribution of truck trips. Trip-based sampling is a useful option for developing freight demand models in countries like India, where CFS practice is absent [12]. Nonetheless, trip-based sampling requires carrier participation, and it fails to identify and model the economic behavior of freight generators because economic behavior is influenced mainly by the commodity attributes [13]. At present, the freight modeling mechanism considers only one dimension of freight demand, whereas freight is characterized by multiple dimensions such as volume, weight, and truck trips. Possibly, more realistic models need to consider more than one dimension to represent freight activities.

The research effort is much needed for developing models considering multiple dimensions of freight activities. This necessity arises from the complicated distribution of urban freight flows that encounter multiple decision-makers, stops, modes, route change, and higher commercial supply-chain value [14, 15]. Multidimensional freight activity also restricts the decision-makers' ability to provide enhanced solutions for seamless freight flow distribution [16]. In this scenario, estimating trip TLD from tonnage TLD and vice-versa will be helpful for urban planners. The conversion from trip TLD to tonnage TLD or vice-versa will enable the analyst to model the trip and tonnage simultaneously at the trip distribution phase. Moreover, if trip/tonnage TLD can be derived from a modeled tonnage/trip TLD, not only will it simplify the complexity of freight demand modeling, but also, it will reduce the burden on survey cost and inventories [13, 17]. Commodity classification is imperative in both trip-based and commodity-based models. It helps to organize the models in a way that complements the decree of land use and policy interventions [18]. Previous studies highlight the use of 'a priori' and 'a posteriori' classification systems in developing freight demand models [19, 20, 21]. It is apparent that class-specific TLDs will be helpful at the trip distribution stage to develop commodity distribution tables. However, accurate estimates of trip TLD and tonnage TLD for different commodity classes within a standard classification system are rare. A research effort is also needed to understand the nature of the relationship between trip TLD and tonnage TLD for different commodity types.

The main objectives of this study are (a) to estimate the distribution function that represents trip TLD and tonnage TLD accurately for several commodity classes; (b) to develop the approximation function that precisely approximates trip TLD from a modeled tonnage TLD function, and vice-versa for each commodity class, and (c) to investigate the relationship between trip TLD and tonnage TLD for different commodity classes.

Research Background

Limited research on TLD for freight transportation is majorly attributed to the complexity arising from a combination of multiple dimensions and decision-makers, who interact in a dynamic ecosystem [10, 13]. Trip length is identified as one of the measures to assess the city logistics scenarios to improve economic and environmental sustainability [3, 22, 23]. TLD models are used to analyze the freight distribution of short-haul and long-haul trips [11]. Ma et al. [17] analyzed 1.3 million GPS records (average trip length, average dwell time, number of trip chains, number of stops per trip chain, etc.) to generate daily truck trip-chaining information. Noekel et al. [24] suggested measuring the freight demand distribution at different distance bands for a better representation of supply characteristics of freight activity. TLD is used as a parameter to investigate the attributes of trip-chaining behavior through a micro-simulation urban freight model [10]. Holguín-veras and Thorson [13] estimated TLD functions for freight movements by classifying trip lengths as intra-city and intercity trips, vehicle type, and commodity groups. However, the commodity groups were limited to manufacturing, mineral, fruit, and vegetable products, leaving a scope to carry out further analysis for the remaining major commodity groups.

Industrial Classification and TLD Function

Industrial or commodity classification is based on the assumption that similar trip/tonnage generation patterns exist among establishments grouped within a classification [19]. However, limited research efforts have been made in the past to estimate TLDs for a standard industrial classification system. Though Holguín-veras and Thorson [13] suggested the gamma function as the appropriate function

to determine freight TLD, passenger transportation studies showed that different functions might be applicable to a different city or trip purpose [1]. Similarly, a single distribution function may not be applicable to all the commodity classes; hence, it is preferable to choose the best fit by comparing different distribution functions based on their root mean squared error (RMSE) values [1, 25] for each commodity class.

Concept of the Approximation Function

The freight demand model system considers three basic measurements: weight or tonnage, trips, and volume. Freight activity can be represented in an improved manner when at least two of these measures can be jointly modeled. Holguín-Veras and Thorson [13] suggested developing an "approximate function" that will be useful in estimating trip TLDs from tonnage TLDs, and thus, simultaneous estimation of both commodity-based and trip-based models will be possible. However, they did not provide any functional form of the approximation function for converting trip TLD to tonnage TLD or vice-versa. Kim et al. [17] proposed an 'intermediate function' to approximate commodity TLD from truck trip TLD. The proposed intermediate function is a quadratic equation where the ratio of trip and commodity TLD is the dependent variable, and trip distance is the independent variable. The function was found to be statistically valid only for miscellaneous and chemical industry items, leaving other items such as agricultural, forestry and fishery, mineral & mining, and other industries unexplored due to the limited availability of observed data. Thus, the literature highlights the need for an approximation function to study the nature and relationship of trip and tonnage TLD characteristics across the major commodity groups. The approximation function can be utilized to get the 'trip-based deterrence function' from a 'tonnage-based deterrence function' and vice-versa. Such a relationship also implies that trip production and attraction can be presumed from tonnage production and attraction. Consequently, tonnage generation (production and attraction) can be obtained from trip generation, which establishes the usefulness of an approximation function to integrate commodity-based and trip-based modeling at the trip distribution stage.

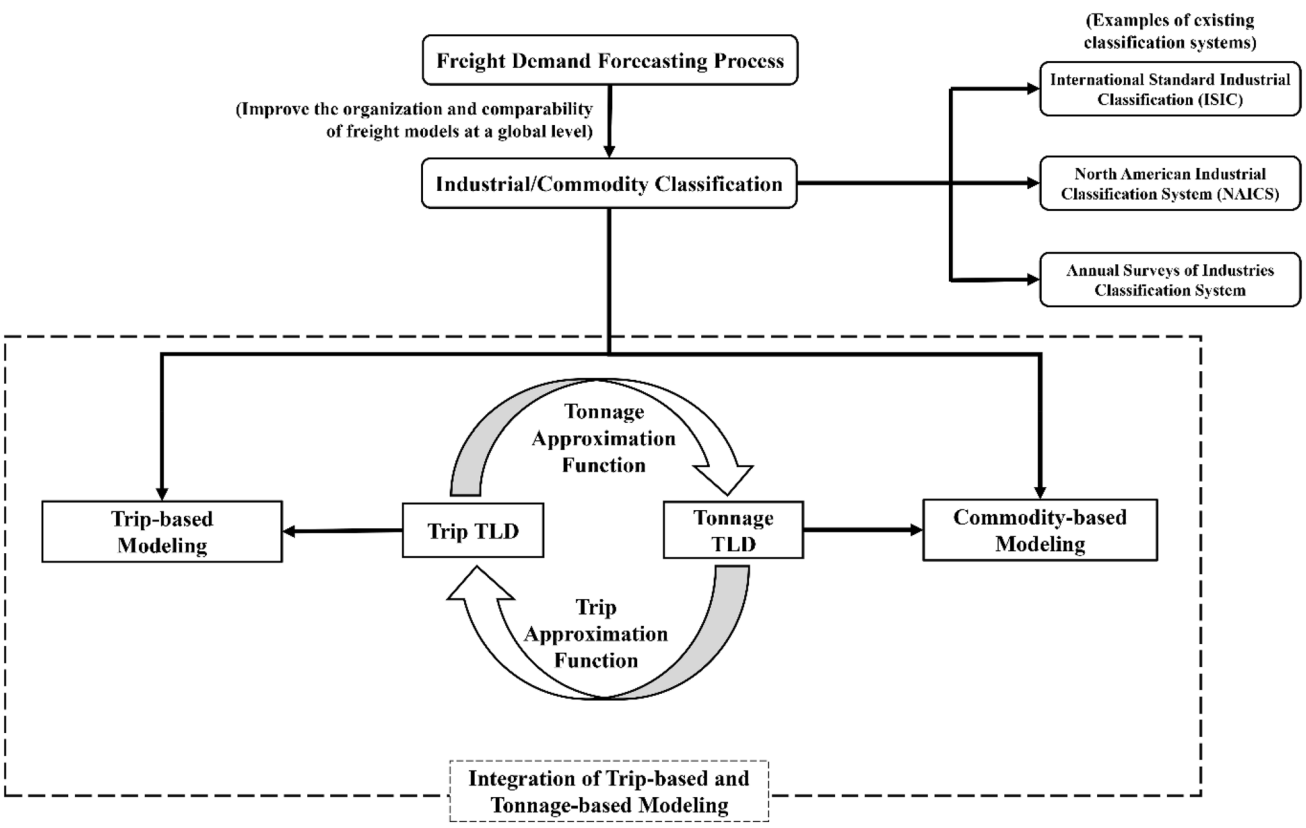

Fig. 1 Trip–tonnage conversion process

Integration of Trip and Tonnage Based Modeling

Trip or tonnage TLD acts as a necessary input to the trip distribution stage through the incorporation of the 'deterrence function,' also known as the friction factor, that represents the disincentive to travel. Trip-based models use truck TLD, and commodity-based models utilize tonnage TLD as the basis of deterrence function in their respective trip distribution modeling stages [10, 13]. Holguín-Veras and Thorson [13] analyzed TLDs for three commodity classes without distinguishing the establishments into any standard industrial classification system. The importance of adopting a standard industrial classification system to result in homogeneous classes of establishments is well validated in recent literature. However, class-specific TLDs have not yet been investigated in the context of integrating trip-based and commodity-based modeling approaches.

The aim is to model the trip and tonnage TLD as well as develop an approximation function for all the major commodity classes within a standard industrial/commodity classification system such as ISIC. The approximation function is to be used as a trip-tonnage conversion factor and vice-versa. The other contribution is the integration of commodity-based and trip-based modeling that can help regional or national planning agencies to produce aggregate estimates of total tons from trips, or vice-versa, as a data-saving exercise. Figure 1 illustrates the use of this trip-tonnage conversion to integrate commodity-based and trip-based modeling for different commodity classes at the trip-distribution stage.

Data

Establishment Based Freight Survey (EBFS) data from Kerala State, India, are used in this research. The EBFS targeted shippers ranging from manufacturing units and wholesalers to raw material production sites in seven cities in Kerala: Cochin, Calicut, Malappuram, Kannur, Palakkad, Thrissur, and Kottayam. The sampling frame is developed using an economic census that provides a list of all industrial establishments in Kerala. Simple random sampling with face-to-face interviews is used for EBFS data collection from logistic managers or establishment owners. The response rate for the survey is 30%. The final sample comprises 432 establishments with complete observations on the total quantity of goods received and produced, the total number of truck trips produced and attracted, employment size, years in business, gross floor area, number of branches, and shipment distance. The EBFS sample is coded using the ISIC system. All the information on the trip and tonnage distance is categorized into ISIC classes (ISIC 10: Food products, ISIC 11: Beverages, tobacco and related products, ISIC 13–14: Textile products, ISIC 16: Wood and wood products, ISIC 17–18: Paper, paper products, and printing, publishing allied industries, ISIC 20–21: Chemicals and chemical products, ISIC 22: Plastic products, ISIC 23: Mineral products, ISIC 24–25: Metal products, ISIC 26–28: Machinery and equipment excluding transport equipment, ISIC 29–30: Manufacturing and transport equipment, ISIC 32: Other products). Only Manufacturing and Other products have 14 observation points on trip and tonnage distance distribution. The remaining samples contain more than 30 observations in each of the categories. A descriptive summary of the data is given in Table 1. More details on survey instrument design, data collection framework, sampling technique, population representativeness, ISIC commodity classifications, and response rates are provided in the literature Pani and Sahu [26, 27].

Methodological Approach

Step 1 Modeling the Distribution Function

The first step is to identify the theoretical probability distribution function (PDF) that best fits the observed trip and tonnage TLD for each ISIC industrial class. The PDF is determined considering the trip length interval as 20 km (the minimum trip length for any commodity class is 23 km). Several PDFs (see Table 2), such as Log-normal, Gamma, Weibull, Normal, and Exponential, are used for analysis. The PDF parameters are estimated using the least-square method [25]. The least-square algorithm to fit distribution parameters is given below:

Least Square Algorithm for fitting Theoretical Distribution Parameters

Identify the $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *for a given dataset.*

- *Linearize the Cumulative distribution function (CDF) by transforming* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *to get the form of* $\boldsymbol{y} = \boldsymbol{mx} + \boldsymbol{c}$
- *Apply linear regression technique to fit a line to the linearized CDF*
- *Extract the parameters of the fitted line*
- *Perform inverse transformation on the parameters of the fitted line*

Return the parameters of the distribution (CDF)

Table 1 Descriptive summary of the data

ISIC	Truck trip length (kilometers)		Tonnage trip length (kilometers)		Sample count	% Sample
	Mean	Standard deviation	Mean	Standard deviation		
ISIC 10 Food products	162.78	93.62	206.01	98.79	98	22.7
ISIC 11 Beverages	179.07	98.21	221.28	106.84	22	5.1
ISIC 13–14 Textile products	130.39	41.85	133.85	42.37	29	6.7
ISIC 16 Wood products	192.38	66.84	201.44	70.71	50	11.6
ISIC 17–18 Paper products	158.33	39.42	166.05	38.57	14	3.2
ISIC 20–21 Chemical products	118.86	37.45	136.46	39.33	44	10.2
ISIC 22 Plastic products	170.36	66.51	213.32	66.81	37	8.6
ISIC 23 Mineral products	198.28	63.33	210.85	63.47	27	6.3
ISIC 24–25 Metal Products	226.26	79.25	243.39	82.93	51	11.8
ISIC 26–28 Machinery	114.33	32.36	122.46	32.79	37	8.6
ISIC 29–30 Transport equipment	172.82	56.07	201.31	44.01	10	2.3
ISIC 32 Others	152.78	37.95	158.95	38.12	13	3
				Total	432	100

Table 2 Functional form of distribution functions used for fitting TLD

Distribution functions	Functional form	Parameters	
Log-normal	$\frac{1}{x\sigma\sqrt{2\pi}}\exp\left(-\frac{(\ln x-\mu)^2}{2\sigma^2}\right)$	μ	σ
Gamma	$\frac{x^{\alpha-1}e^{\frac{-x}{\beta}}}{\beta^{\alpha}\Gamma(\alpha)}$	α	β
Weibull	$\frac{k}{\lambda^k}\left(\frac{x}{\lambda}\right)^{k-1}\exp-\left(\frac{x}{\lambda}\right)^k$	λ	k
Normal	$\frac{1}{\sigma\sqrt{2\pi}}\exp-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2$	μ	σ
Exponential	$\lambda\exp-\lambda x$	λ	

In the present case, x represents the trip distance (length), and y represents the probability (relative frequency of a trip). Once the parameters are estimated for each theoretical distribution, RMSE values (as given in Eq. 1) are used to find the best fit function. In Eq. 1, n represents the sample size.

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(y_{\mathrm{Observed}} - y_{\mathrm{Modeled}}\right)^2}{n}} \tag{1}$$

Step 2 Goodness of Fit Test for Modeled TLD

Using the best fit theoretical PDF, the trip and tonnage TLDs are estimated at a 20-km interval, and the estimated TLDs are termed as 'modeled TLDs.' Kolmogorov Smirnov (*K–S*), and χ^2 tests are used to check the statistical validity (i.e., the estimated probability matches the observed probability) at 99% confidence level. The null hypothesis in both tests states that there is no significant difference between the observed and the modeled TLDs. In this study, the Kolmogorov–Smirnov test compares the empirical cumulative distribution function (ECDF) obtained from the observed trip frequency (y_{Observed}) and the cumulative distribution function of the best fit theoretical trip frequency (y_{Modeled}) obtained from Step 1. The K–S test calculates the test statistic, D_{Test} that represents the upper extreme among absolute value differences between theoretical CDF and ECDF. The hypothesis on the similarity of observed and modeled TLD is rejected if $D_{\mathrm{Test}} > D_{\mathrm{critical}}$ (the critical value) where D_{Critical} represents the critical value of the test statistic at a 99% confidence level. Similarly, in χ^2 test, the test statistic (χ^2_{test}) is calculated (as shown in Eq. 2), and the null hypothesis is rejected, if $\chi^2_{\mathrm{test}} > \chi^2_{\mathrm{critical}}$, i.e., test statistic at 99% confidence level (alternatively, p value ≤ 0.01).

$$\chi^2_{\text{test}} = \sum_{i=1}^{k} \left(\frac{\left(y_i^{\text{Observed}} - y_i^{\text{Modeled}}\right)^2}{y_i^{\text{Modeled}}} \right) \tag{2}$$

Here k is the number of 20-km intervals in the data.

Step 3 Estimation of Approximation Function

The rationale behind developing an approximation function lies in the hypothesis that if the theoretical truck TLD function is known, it can approximate tonnage TLD and vice-versa. The mathematical basis for approximating trip and tonnage TLD using theoretical TLDs is given in Eqs. 3 and 4.

$$f\left(x_{\text{Approximated tonnage TLD}}\right) = f\left(x_{\text{Theoretical truck TLD function}}\right) \times f\left(x_{\text{Tonnage approximation function}}\right) \tag{3}$$

$$f\left(x_{\text{Approximated truck TLD}}\right) = f\left(x_{\text{Theoretical tonnage TLD function}}\right) \times f\left(x_{\text{Truck approximation function}}\right) \tag{4}$$

The next task lies in finding the suitable expressions for $f\left(x_{\text{Tonnage approximation function}}\right)$ and $f\left(x_{\text{Truck approximation function}}\right)$ in terms of trip distance since the theoretical distribution functions are also in terms of trip distance. Under the assumption that the theoretical TLDs match closely with observed TLDs, as follows:

$$\text{Approximated Tonnage TLD} = \text{Theoretical Truck TLD} \times \frac{\text{Observed Tonnage TLD}}{\text{Observed Truck TLD}} \tag{5}$$

$$\text{Approximated Truck TLD} = \text{Theoretical Tonnage TLD} \times \frac{\text{Observed Truck TLD}}{\text{Observed Tonnage TLD}} \tag{6}$$

Equations 5–6 are logically intuitive. If the distribution fits are reliable, then the theoretical distribution matches closely with the observed data. Consequently, the approximated TLDs also match well the observed data. Based on the discussion so far, it appears logical if it can express the ratios $\left(\frac{\text{Observed Tonnage TLD}}{\text{Observed Truck TLD}}\right)$ and $\left(\frac{\text{Observed Truck TLD}}{\text{Observed Tonnage TLD}}\right)$ as a function of trip distance. To develop such a function, it took the help of three types of equations, namely quadratic, cubic, and log transform equations. The functional forms of the three equations are shown in Eqs. 7–9.

$$\text{Quadratic: } f(x) = \alpha + \beta_1 x + \beta_2 x^2 \tag{7}$$

$$\text{Cubic: } f(x) = \alpha + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 \tag{8}$$

$$\text{Log Transformation: } f(x) = \alpha + \log x \tag{9}$$

Here, α is the intercept, and x is the desired independent variable, i.e., trip distance in this case. The idea to choose these three specific forms of approximation functions comes from literature. Literature suggests that probability distribution functions can be approximated using polynomials [28]. Additionally, the log transformation equation contains the exponential term present in all the distribution functions and has been used in previous studies dealing with similar problems [17]. Hence, the three functional forms appeared intuitive. The authors did not consider higher order polynomials as cubic order was giving a reasonably good fit, and using a higher order polynomial may lead to overfitting [29]. Finally, the quest to find a suitable approximation function ends by regressing the ratio of observed TLDs on trip distance. For the tonnage approximation function, the ratio of observed tonnage distribution to observed truck-trip distribution $\left(\frac{\text{Observed Tonnage TLD}}{\text{Observed Truck TLD}}\right)$ is taken as the dependent variable. In the case of trip approximation function, the dependent variable is the ratio of observed truck-trip distribution to observed tonnage distribution $\left(\frac{\text{Observed Truck TLD}}{\text{Observed Tonnage TLD}}\right)$. Trip distance remains the independent variable in both cases. The three functions are estimated and compared based on the statistical significance of coefficients and R-square (R^2) values to get the best fit approximation function.

Step 4 Goodness of Fit Test for Approximated TLD

After the trip and tonnage TLDs are approximated, the goodness of fit test is conducted to check if the approximated trip and tonnage TLD match their respective observed TLD. Again, the test statistic (χ^2_{test}) is calculated (as shown in Eq. 10), and the null hypothesis is rejected, if $\chi^2_{\text{test}} > \chi^2_{\text{critical}}$, i.e., test statistics at a 99% confidence level (alternatively, p-value ≤ 0.01).

$$\chi^2_{\text{test}} = \sum_{i=1}^{k} \left(\frac{\left(y_i^{\text{Observed}} - y_i^{\text{Approximated}}\right)^2}{y_i^{\text{Approximated}}} \right) \tag{10}$$

Here k is the number of 20-km intervals in the data.

Results Analysis and Discussion

Preliminary Investigation

The TLDs for the entire data set are plotted in Fig. 2 to get an initial understanding of the overall nature of trip and

tonnage distributions. From the plot, the following discussions are inferred: (a) the peak of trip TLD and tonnage TLD are visibly different, and a similar inference is reported by Holguín-veras and Thorson [13], (b) the peak for trip TLD lies in the 100–120 km (km) range whereas, the peak for tonnage TLD is in 160–200 km range, and (c) contrary to the existing literature [13], the difference between trip and tonnage TLD does not seem to disappear with increasing distance. This initial understanding ascertains the complexity of freight transport by justifying that commodity (tons) and truck trips are the two significantly different units of analysis, thus, establishing the need for an approximation function that integrates these two units. A majority of truck-trips and tonnage are long haul (trip distance is ≥ 100 km) and can be categorized as intercity trips. The trend justifies the economic and geographic land use pattern of the study area [30]. Major seaports serve four of the seven cities in Kerala with a higher degree of industrialization. The cities are also comprised of major agricultural areas, highly populated residential zones, and special economic zones that add to the spikes in the TLDs. The results obtained from fitting distribution (discussed in the later sections), also hint towards similar findings. The preliminary understanding of overall trip and tonnage TLD patterns reflect the special land uses and can assist planners to successfully incorporate their effects while developing freight demand models [31]. The overall trip and tonnage TLD also govern the fleet ownership decisions, freight transport expenditures, and consequently possible resulting emissions. Hence, this preliminary understanding may also in developing effective logistics plans and marketing strategies.

Modeled Trip and Tonnage TLD

Table 3 summarizes the RMSE values for the distribution functions for both trip and tonnage distributions in each ISIC class. The highlighted cells with the lowest RMSE values are the best-fit PDFs. The trip TLD for ISIC 10 and 32 follows Gamma distribution, whereas Weibull distribution is the best fit for ISIC 11, 13–14, 22, 26–28 and 29–30.

The best fit function for trip TLD in ISIC 16, 20–21, 23 and 24–25 is Log-normal. ISIC 17–18 is the only commodity group that follows a normal distribution for trip TLD. The tonnage TLD for ISIC 10, 13–14, 20–21 and 22 is best represented by a Weibull distribution, whereas; Gamma distribution is appropriate for ISIC 17–18, 26–28 and 32. A Lognormal distribution best represents the tonnage TLD for ISIC 16, 23, 24–25 and 29–30. The Gamma, Weibull, and Lognormal distributions do not apply to trip or tonnage TLD for some commodities because of smaller sample sizes. The discussions further add to the freight literature as no specific distribution is found to predominantly represent the trip or tonnage TLD. This inference will be of particular interest to the urban planners and

Fig. 2 Trip and tonnage TLD

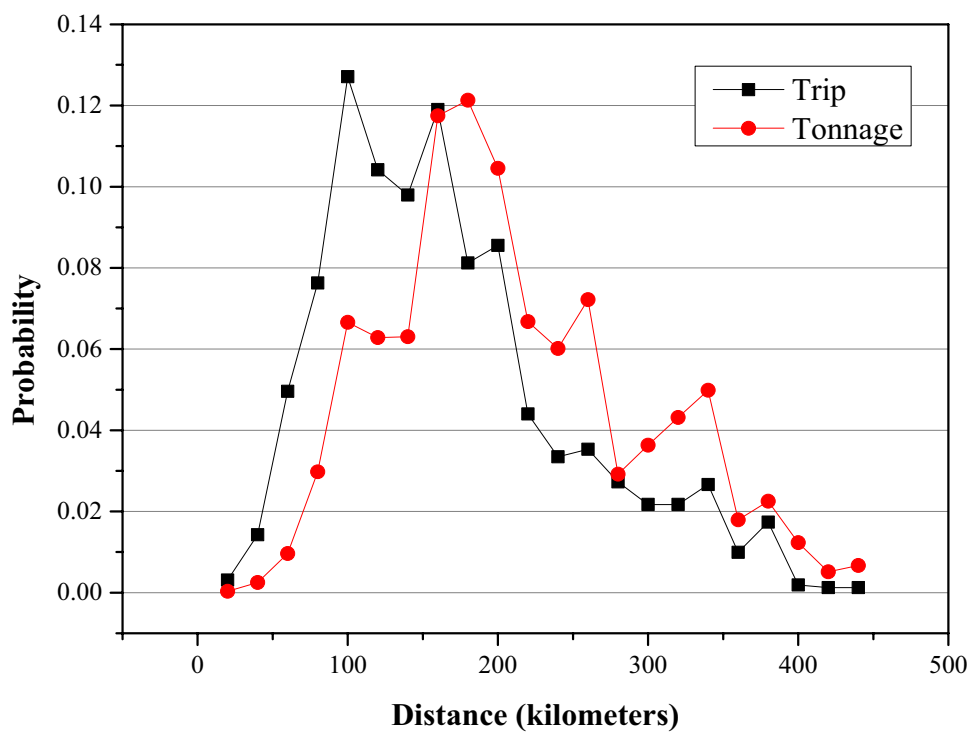

Table 3 RMSE values of modeled distribution functions

Distributions		Log-normal	Gamma	Weibull	Normal	Exponential
ISIC 10 Food products	Trip	1.012E−03	**9.64E−04**	1.05E−03	1.319E−03	1.07E−03
	Tonnage	NA	1.665E−03	**1.63E−03**	1.666E−03	1.79E−03
ISIC 11 Beverages	Trip	2.66E−03	2.65E−04	**2.63E−03**	2.91E−03	2.64E−03
	Tonnage	NA	**3.893E−03**	3.90E−03	3.95E−03	3.894E−03
ISIC 13–14 Textile products	Trip	2.902E−03	2.887E−04	**2.853E−03**	3.207E−03	3.25E-03
	Tonnage	3.316E−03	3.289E−03	**3.275E−03**	3.582E−03	3.682E−03
ISIC 16 Wood products	Trip	**1.196E−03**	1.20E−04	1.34E−03	1.21E−03	1.425E−03
	Tonnage	**1.39E−03**	1.58E−03	1.67E−03	1.62E−03	1.49E−03
ISIC 17–18 Paper products	Trip	3.90E−03	NA	NA	**3.84E−03**	5.1E−03
	Tonnage	3.68E−03	**3.07E−03**	4.05E−03	4.29E−03	3.72E−03
ISIC 20–21 Chemical products	Trip	**1.326E−03**	1.337E−03	1.39E−03	1.78E−03	1.88E−03
	Tonnage	NA	NA	**2.279E−03**	2.37E−03	2.94E−03
ISIC 22 Plastic products	Trip	1.49E−03	1.47E−03	**1.46E−03**	1.78E−03	1.66E−03
	Tonnage	NA	NA	**2.32E−03**	2.35E−03	2.80E−03
ISIC 23 Mineral products	Trip	**2.44E−03**	2.55E−03	2.82E−03	2.81E−03	2.73E−03
	Tonnage	**2.46E−03**	2.98E−03	2.70E−03	3.42E−03	3.05E−03
ISIC 24–25 Metal Products	Trip	**2.44E−03**	2.55E−03	2.82E−03	2.81E−03	2.73E−03
	Tonnage	**2.46E−03**	2.98E−03	2.70E−03	3.42E−03	3.05E−03
ISIC 26–28 Machinery	Trip	2.49E−03	2.46E−03	**2.44E−03**	2.63E−03	3.40E−03
	Tonnage	2.40E−03	**2.28E−03**	6.20E−03	2.71E−03	2.60E−03
ISIC 29–30 Transport equipment	Trip	NA	NA	**5.10E−03**	5.12E−03	5.40E−03
	Tonnage	**5.80E−03**	5.90E−03	6.70E−03	6.60E−03	6.30E−03
ISIC 32 Others	Trip	3.45E−03	**3.38E−03**	4.45E−03	4.49E−03	5.60E−03
	Tonnage	3.78E−03	**3.501E−03**	4.307E−03	4.378E−03	5.70E−03

NA not applicable

modelers about the concerning errors in results if a single distribution function is assumed for all the commodity classes. The goodness of fit test results, as shown in Table 4, gives statistical evidence that all the modeled TLDs match closely to the observed TLDs. The separate PDFs to estimate the trip and tonnage TLDs for each commodity group are also given in Table 4. The results from the *K*–*S* test show that D_{Test} does not exceed D_{Critical} for any of the cases. The results from χ^2 test indicate p value ≥ 0.05 for all commodity classes. Thus, the null hypothesis (i.e., modeled TLD and observed TLD are not different) cannot be rejected.

Approximation Function

The statistics of trip and tonnage approximation functions for all the commodity classes are summarized in Table 5. The trip and tonnage approximation functions for ISIC 11, 16, 23, 24–25, 26–28, 29–30 and 32 are represented by a log transformation equation. The trip and tonnage approximation functions for ISIC 10, 13–14 and 22 are quadratic. The trip approximation function is quadratic for ISIC 17–18 and cubic for ISIC 20–21. The tonnage approximation function is cubic for ISIC 17–18 and quadratic for ISIC 20–21. This finding suggests that the functional form of the approximation function ranges from cubic and quadratic to log

Table 4 Goodness of fit test statistics for modeled TLD (null hypothesis: observed TLD and modeled TLD are not different)

		Distribution function	Kolmogorov Smirnov test		χ^2 test				Distribution function	Kolmogorov Smirnov test		χ^2 test	
			D_{Test}	D_{Critical}	χ^2	p value				D_{Test}	D_{Critical}	χ^2	p value
ISIC 10	Trip	$\frac{x^{(0.311)}e^{\frac{-x}{118.99}}}{118.99^{1.311}\Gamma(1.311)}$	0.082	0.352	0.007	0.999	ISIC 22	Trip	$\frac{1.39}{120.52^{1.39}}x^{(0.39)}e^{-\left(\frac{x}{120.52}\right)^{1.39}}$	0.051	0.391	0.013	0.999
	Tonnage	$\frac{1.766}{260.66^{1.766}}x^{(0.766)}e^{-\left(\frac{x}{260.66}\right)^{1.766}}$	0.154	0.352	0.023	0.999		Tonnage	$\frac{6.64}{468.43^{6.64}}x^{(5.64)}e^{-\left(\frac{x}{468.43}\right)^{6.64}}$	0.111	0.391	0.031	0.999
ISIC 11	Trip	$\frac{0.966}{174.57^{0.966}}x^{(-0.034)}e^{-\left(\frac{x}{174.57}\right)^{0.966}}$	0.166	0.370	0.061	0.999	ISIC 23	Trip	$\frac{e^{-\left(\frac{(\ln(x)-100.72)^2}{2\times83.56^2}\right)}}{\sqrt{2\pi}\times83.56\times x}$	0.125	0.432	0.031	0.999
	Tonnage	$\frac{x^{(-0.01)}e^{\frac{-x}{232.49}}}{232.49^{0.99}\Gamma(0.99)}$	0.246	0.370	0.122	0.999		Tonnage	$\frac{e^{-\left(\frac{(\ln(x)-104.26)^2}{2\times240.47^2}\right)}}{\sqrt{2\pi}\times240.47\times x}$	0.219	0.432	0.035	0.999
ISIC 13–14	Trip	$\frac{1.319}{84.056^{1.319}}x^{(0.319)}e^{-\left(\frac{x}{84.056}\right)^{1.319}}$	0.081	0.467	0.035	0.999	ISIC 24–25	Trip	$\frac{x^{(2.95)}e^{\frac{-x}{44.37}}}{44.37^{3.95}\Gamma(3.95)}$	0.081	0.467	0.035	0.999
	Tonnage	$\frac{1.36}{88.40^{1.36}}x^{(0.36)}e^{-\left(\frac{x}{88.40}\right)^{1.36}}$	0.098	0.467	0.046	0.999		Tonnage	$\frac{1.58}{172.53^{1.58}}x^{(0.58)}e^{-\left(\frac{x}{172.53}\right)^{1.58}}$	0.098	0.467	0.046	0.999
ISIC 16	Trip	$\frac{e^{-\left(\frac{(\ln(x)-321.93)^2}{2\times48.81^2}\right)}}{\sqrt{2\pi}\times48.81\times x}$	0.082	0.344	0.011	0.999	ISIC 26–28	Trip	$\frac{1.97}{77.32^{1.97}}x^{(0.97)}e^{-\left(\frac{x}{77.32}\right)^{1.97}}$	0.060	0.513	0.011	0.999
	Tonnage	$\frac{e^{-\left(\frac{(\ln(x)-108.23)^2}{2\times120.51^2}\right)}}{\sqrt{2\pi}\times120.51\times x}$	0.105	0.344	0.017	0.999		Tonnage	$\frac{x^{(0.64)}e^{\frac{-x}{39.28}}}{39.28^{1.64}\Gamma(1.64)}$	0.017	0.513	0.017	0.999
ISIC 17–18	Trip	$\frac{e^{-\left(\frac{(x-173.27)^2}{2\times18.78^2}\right)}}{\sqrt{2\pi}\times18.78}$	0.182	0.449	0.070	0.999	ISIC 29–30	Trip	$\frac{2.95}{161.19^{2.95}}x^{(1.95)}e^{-\left(\frac{x}{161.19}\right)^{2.95}}$	0.130	0.488	0.088	0.999
	Tonnage	$\frac{x^{(3.52)}e^{\frac{-x}{3.38}}}{3.38^{4.52}\Gamma(4.52)}$	0.378	0.449	0.065	0.999		Tonnage	$\frac{e^{-\left(\frac{(\ln(x)-121.5)^2}{2\times5.76^2}\right)}}{\sqrt{2\pi}\times5.76\times x}$	0.344	0.488	0.161	0.999
ISIC 20–21	Trip	$\frac{e^{-\left(\frac{(\ln(x)-108.38)^2}{2\times49.85^2}\right)}}{\sqrt{2\pi}\times49.85\times x}$	0.078	0.488	0.006	0.999	ISIC 32	Trip	$\frac{x^{(-0.004)}e^{\frac{-x}{67.32}}}{67.32^{0.996}\Gamma(0.996)}$	0.129	0.541	0.035	0.999
	Tonnage	$\frac{4.34}{195.16^{4.34}}x^{(3.34)}e^{-\left(\frac{x}{195.16}\right)^{4.34}}$	0.112	0.488	0.017	0.999		Tonnage	$\frac{x^{(0.004)}e^{\frac{-x}{70.61}}}{70.61^{1.004}\Gamma(1.004)}$	0.139	0.541	0.036	0.999

x is the trip distance in kilometer

Table 5 Statistics of approximation function

	Function type	Variables	Coefficients	t-stat	p value	R^2		Function type	Variables	Coefficients	t stat	p value	R^2
ISIC 10 Food products							ISIC 22 Plastic products						
Tonnage	Quadratic	Intercept	8.060E−01	4.186	0.001	94.79	Tonnage	Quadratic	Intercept	7.837E−01	2.483	0.047	98.37
		x	− 5.512E−03	− 2.360	0.039				x	− 1.051E−02	− 2.726	0.034	
		x^2	2.816E−05	4.931	0.0005				x^2	5.861E−05	5.534	0.001	
Trip	Quadratic	Intercept	2.309	12.830	2.29E−08	89.23	Trip	Quadratic	Intercept	6.749	6.547	0.0001	76.32
		x	− 9.635E−03	− 4.816	0.0004				x	− 4.526E−02	− 3.987	0.003	
		x^2	1.338E−05	2.847	0.14				x^2	8.124E−05	2.851	0.019	
ISIC 11 Beverage products							ISIC 23 Mineral products						
Tonnage	Log Transformation	Intercept	− 2.6073	− 3.645	0.006	75.23	Tonnage	Log Transformation	Intercept	− 2.435	− 4.404	0.0022	82.75
		ln x	0.7056	4.941	0.001				ln x	0.6465	6.194	0.0002	
Trip	Log Transformation	Intercept	10.127	15.53	1.11E−06	96.07	Trip	Log Transformation	Intercept	4.983	9.186	7.22E−06	85.16
		ln x	− 1.6892	− 13.09	3.55E−06				ln x	− 0.737	− 7.187	5.15E−05	
ISIC 13–14 Textiles products							ISIC 24–25 Metal products						
Tonnage	Quadratic	Intercept	1.104	9.451	0.002	92.93	Tonnage	Log Transformation	Intercept	− 3.2585	− 5.105	0.0002	75.84
		x	− 4.131E−03	2.573	0.082				ln x	0.8065	0.119	2.02E−05	
		x^2	1.936E−05	− 3.386	0.042								
Trip	Quadratic	Intercept	9.092E−01	15.195	0.002	95.74	Trip	Log Transformation	Intercept	4.2103	7.556	1.12E−05	75.57
		x	4.102E−03	− 3.432	0.082				ln x	− 0.5972	− 5.834	0.0001	
		x^2	− 1.933E−05	4.492	0.042								
ISIC 17–18 Wood products							ISIC 26–28 Machinery						
Tonnage	Cubic	Intercept	2.457	2.648	0.033	75.35	Tonnage	Log Transformation	Intercept	− 3.5882	− 3.721	0.007	76.44
		x	− 2.692E−02	− 2.105	0.073				ln x	0.9776	4.766	0.002	
		x^2	1.343E−04	2.510	0.040								
		x^3	− 1.851E−07	2.696	0.031								
Trip	Quadratic	Intercept	3.104	5.308	0.0018	76.32	Trip	Log Transformation	Intercept	13.195	4.905	0.001	72.55
		x	− 1.426E−02	− 3.173	0.019				ln x	− 2.461	− 4.301	0.003	
		x^2	2.124E−05	2.647	0.038								
ISIC 16 Paper products							ISIC 29–30 Transport equipment						
Tonnage	Log Transformation	Intercept	− 2.8487	− 3.357	0.028	84.25	Tonnage	Log Transformation	Intercept	− 7.509	− 4.097	0.014	84.01
		ln x	0.7731	4.626	0.009				ln x	1.6485	4.584	0.010	
Trip	Log Transformation	Intercept	5.0760	5.326	0.005	82.08	Trip	Log Transformation	Intercept	24.223	4.817	0.008	83.03
		ln x	− 0.8035	− 4.280	0.012				ln x	− 4.365	− 4.424	0.011	
ISIC 20–21 Chemical products							ISISC 32 Others						
Tonnage	Quadratic	Intercept	1.104	9.451	0.002	92.93	Tonnage	Log Transformation	Intercept	− 2.305	− 2.261	0.086	72.88
		x	− 4.131E−03	2.573	0.082				ln x	0.6568	3.279	0.031	
		x^2	1.936E−05	− 3.386	0.042								
Trip	Cubic	Intercept	1.677E+01	7.321	0.005	98.59	Trip	Log Transformation	Intercept	3.837	9.401	0.002	83.03

Table 5 (continued)

Function type	Variables	Coefficients	*t*-stat	*p* value	R^2	Function type	Variables	Coefficients	*t* stat	*p* value	R^2
	x	− 3.101E−01	− 5.009	0.015			ln x	− 0.551	− 6.829	0.006	
	x^2	2.038E−03	3.994	0.028							
	x^3	− 4.493E−06	− 3.437	0.041							

transformation. However, the existence of the log transformation equation as an approximation function for the majority of the ISIC categories is evident. A primary reason for the emergence of log transformation as the dominant functional form can be attributed to the theoretical distribution functions. Majority of them are special cases of the exponential distribution and the log transformation equation offers the exponential term. Table 6 gives statistical evidence that approximated trip and tonnage TLDs match closely to the observed TLDs. Hence, validates the assumption behind the development of the approximation function as truck-to-tonnage or tonnage-to-truck TLD conversion. The practical utilization of the approximation function is also rather simple. For example, in short-term planning, the planners need not collect both the latest tonnage and truck TLDs if they have historical data on truck and tonnage TLDs. They can only collect the latest truck TLD and estimate the theoretical distribution function. After developing a suitable approximation function from the historical data, they can simply obtain the approximated latest tonnage TLD based on the proposed methodology in this paper. Such an exercise is not only crucial in saving resources on survey instrument design and data collection but integrates commodity-based and trip-based modeling at the trip distribution stage.

TLD Characteristics Across Commodity Classes

The observed, modeled, and approximated TLD plots for different commodity classes are shown in Figs. 3, 4, 5 and 6. The plots for ISIC 17–18, 20–21, 29–30 and 32 are not presented due to space constraints; however, discussions on their trip and tonnage TLD characteristics are provided. For ISIC 10 (food product), the trip lengths of the observed, modeled, and approximated trips are within a distance of 260 km, and the tonnage length is within a distance of 340 km. In ISIC 11 class (beverage), more than 80% of trips and 75% of tonnage lie within 260 km. Trip and tonnage distribution patterns for ISIC 11 are similar but so is not the case for ISIC 10. The observed tonnage TLD peak in ISIC 10 occurs at 340 km, and for ISIC 11, it happens at 360 km. The appearance of such peaks at a longer distance is attributed to some distinctive land use patterns and the influence of port cities in-housing a significant proportion of ISIC 10 and 11 establishments (e.g., 20% of ISIC 10 and 11 establishments belong to port city Cochin). Establishments in port cities produce more freight compared to other cities [32]. The peaks of observed, modeled, and approximated trip TLDs, as well as tonnage TLDs for ISIC 13–14 (textile), are within 120 km. The smaller trip length of this commodity class is attributed to their absence in port cities like Cochin. The bulk of trip and tonnage for ISIC 16 (wood product) are distributed within 220 km. The trip and tonnage distributions are more similar in pattern than the

Table 6 Goodness of fit test of approximated TLD (null hypothesis: observed TLD and approximated TLD are not different)

Commodity type		χ^2	p value	Suitability
ISIC 10 Food products	Tonnage	0.0231	0.999	Does not reject the null hypothesis
	Trip	0.0249	0.999	Does not reject the null hypothesis
ISIC 11 Beverages	Tonnage	0.1213	0.999	Does not reject the null hypothesis
	Trip	0.0684	0.999	Does not reject the null hypothesis
ISIC 13–14 Textile products	Tonnage	0.1111	0.999	Does not reject the null hypothesis
	Trip	0.0971	0.999	Does not reject the null hypothesis
ISIC 16 Wood products	Tonnage	0.0216	0.999	Does not reject the null hypothesis
	Trip	0.0171	0.999	Does not reject the null hypothesis
ISIC 17–18 Paper products	Tonnage	0.1037	0.999	Does not reject the null hypothesis
	Trip	0.0723	0.999	Does not reject the null hypothesis
ISIC 20–21 Chemical products	Tonnage	0.0255	0.999	Does not reject the null hypothesis
	Trip	0.0243	0.999	Does not reject the null hypothesis
ISIC 22 Plastic products	Tonnage	0.0295	0.999	Does not reject the null hypothesis
	Trip	0.0224	0.999	Does not reject the null hypothesis
ISIC 23 Mineral products	Tonnage	0.0432	0.999	Does not reject the null hypothesis
	Trip	0.0280	0.999	Does not reject the null hypothesis
ISIC 24–25 Metal Products	Tonnage	0.0118	0.999	Does not reject the null hypothesis
	Trip	0.0114	0.999	Does not reject the null hypothesis
ISIC 26–28 Machinery	Tonnage	0.0205	0.999	Does not reject the null hypothesis
	Trip	0.0589	0.999	Does not reject the null hypothesis
ISIC 29–30 Transport equipment	Tonnage	0.1805	0.999	Does not reject the null hypothesis
	Trip	0.2106	0.999	Does not reject the null hypothesis
ISIC 32 Others	Tonnage	0.0328	0.999	Does not reject the null hypothesis
	Trip	0.0367	0.999	Does not reject the null hypothesis

previously discussed commodity classes. The TLDs for ISIC 17–18 (paper product) are comparable to ISIC 16, whereas trip and tonnage characteristics in ISIC 20–21 (chemical product) are distinct. ISIC 24–25 (metal product) has similar trip and tonnage TLD characteristics, but the frequency is higher with peaks at 240 km, and trip lengths are spreading to 440 km. This pattern substantiates the influence of ports that facilitates the exchange of metal alloys within and outside the country [33]. ISIC 26–28 resembles ISIC 24–25 but with shorter trip distributions. The TLD peak for ISIC 29–30 is 200 km, with the longest distribution of trip length at 260 km. The trip and tonnage TLD peaks for ISIC 32 lie at 140 km with a maximum trip length of 220 km.

An overall inspection suggests that trip and tonnage TLD characteristics for each commodity are different except for the classes ISIC 16 and 23. Each commodity class has a different spatial distribution for truck-trips and tonnages, which necessitates an efficient logistic plan [34, 35]. The use of the approximation function to convert commodity tonnage to truck trips and truck trips to commodity tonnage would be pivotal in preparing the practical logistic plan for different commodity classes. TLDs also give an extent of the distance to which demand exists for each commodity and represent the land use pattern. This discussion is made based on the visible influence of ports on TLD characteristics. The difference between observed, modeled, and approximated TLDs tends to decrease as trip distance increases. Such observations comply with previous studies that showed spatial and land use characteristics have a considerable effect on zonal level variability in freight trips generated [19, 32,

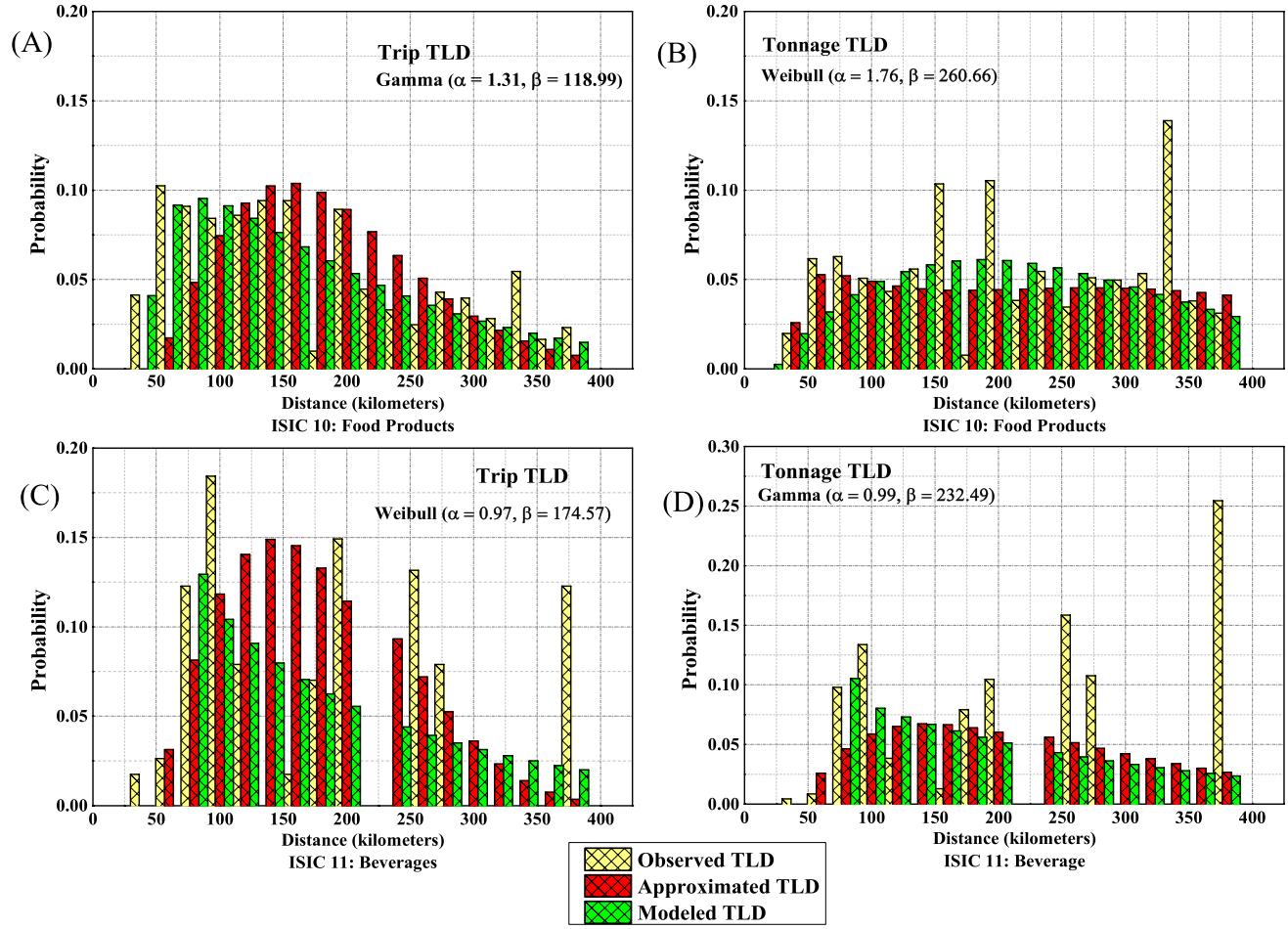

Fig. 3 Trip and tonnage TLD for food products (**a**, **b**) and beverage products (**c**, **d**)

36–38]. Commodity types such as ISIC 10 and 11 could be grouped within a single cluster due to the similar nature of trip and tonnage TLDs. Similarly, ISIC 16 and 17–18 can be arranged in one class, whereas ISISC 13–14 and 20–21 can be grouped in another. Such consideration will reduce the modeling burden through an aggregation scheme for commodities.

Majority of the distribution functions used in the paper are positively skewed, suggesting that most truck and tonnage trip frequencies will be clustered around the left tail of the distribution while the right tail of the distribution is longer. The answer on which location in the left tail will witness the peak frequency and what is the rate by which frequency will fall along the long tail will depend on parameters (obtained from the least square) of the distribution function. For example, both truck and tonnage TLD for ISIC 32 shows Gamma as the best fit. However, based on the α and β parameters, the authors can suggest that the rate at which tonnage frequency will fall along the distance is higher than the rate at which trip frequency will fall. This is attributed to the higher value of β for tonnage TLD as compared to truck TLD. Similar data-dependent frequentist inferences can be obtained from cases that showed Weibull, Lognormal, and Exponential functions as best-fitted distributions. In the end, the type of distribution also reflects upon the haul length variation of the type/class of commodity. All decisions based on haul length variation, such as volume of expenditure, shipment size, and fleet choice, will then be a function of the type and parameters of the distribution function. There lies the beauty of understanding TLD patterns and the importance of TLD functions in logistics planning and decision-making.

Research Application and Limitations

This paper contributes to the quest for enhancing the urban freight models that can jointly model commodity tonnage and truck-trip. The integration of commodity-based models and trip-based models would represent a more realistic scenario of freight activity. The use of the approximation function would provide a basis for urban planners and

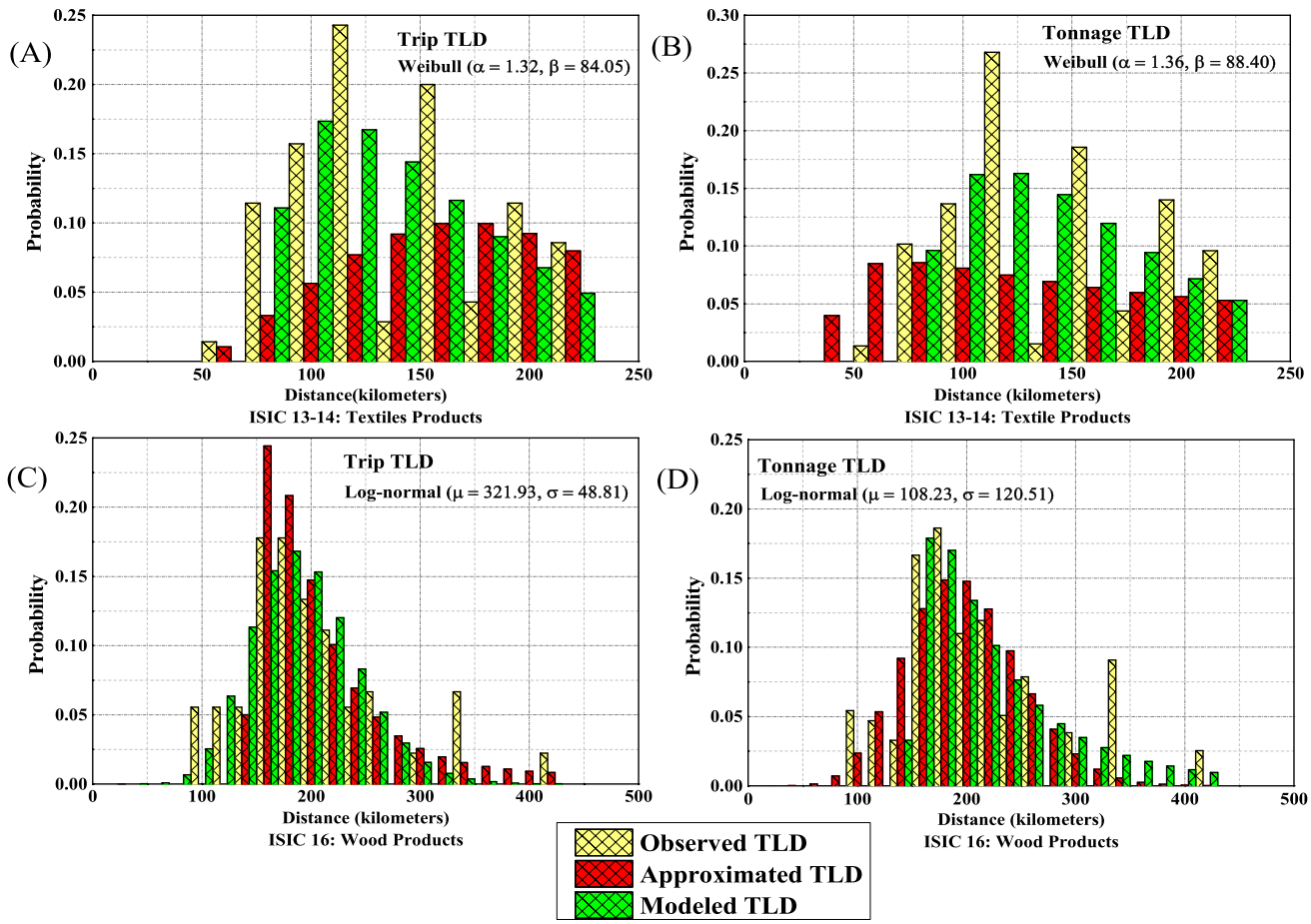

Fig. 4 Trip and tonnage TLD for textile products (**a**, **b**) and wood products (**c**, **d**)

policymakers to model both trips and tons jointly at the trip distribution stage. Given the absence of CFS practice in developing countries, including India, the approximation functions will help in reducing the reporting burden of urban freight surveys leading to a significant saving in resources for data collection. The concept of approximation function can also be considered as a surrogate measure for the tonnage-trip conversion factor known as the truck equivalency factor. Along with that, the paper provides detailed insights into the tonnage and truck-trip length distribution of several commodity classes, which is missing in the literature. Another possible application of this research is that vehicle emission can be approximated by reconstructing the TLD curves.

Like any other research, this research also has certain limitations. For example, the considerable variability in shipment distances both within the commodity classes (as shown in Figs. 3 through 6) as well as by the geography limits the transferability of results that depend on both sample size and regional geography. The results may help other regional studies to produce regional aggregate estimates of total tons moved from trips taken, or vice-versa, as a data-saving exercise: but the region-specific empirical results may not be transferrable elsewhere. Transferability analysis of the approximate functions and the TLDs with larger samples remain as the further scope of this research.

Conclusions

This research is built on the need for integration of trip-based modeling and commodity-based modeling at the trip distribution stage for different commodity groups to provide an enhanced understanding of freight movements. An extensive literature review suggested that limited research efforts are made to understand the relationship between trip and tonnage TLD. The proposition for this research is that if a theoretical distribution function for trip TLD is known, then tonnage TLD can be obtained by multiplying a tonnage approximation function with it and vice-versa. This proposition is tested on a dataset consisting of 432 establishments' trip and tonnage length information. The establishments are classified into 12 ISIC classes, and the TLDs are fitted with five probability distribution functions (PDFs):

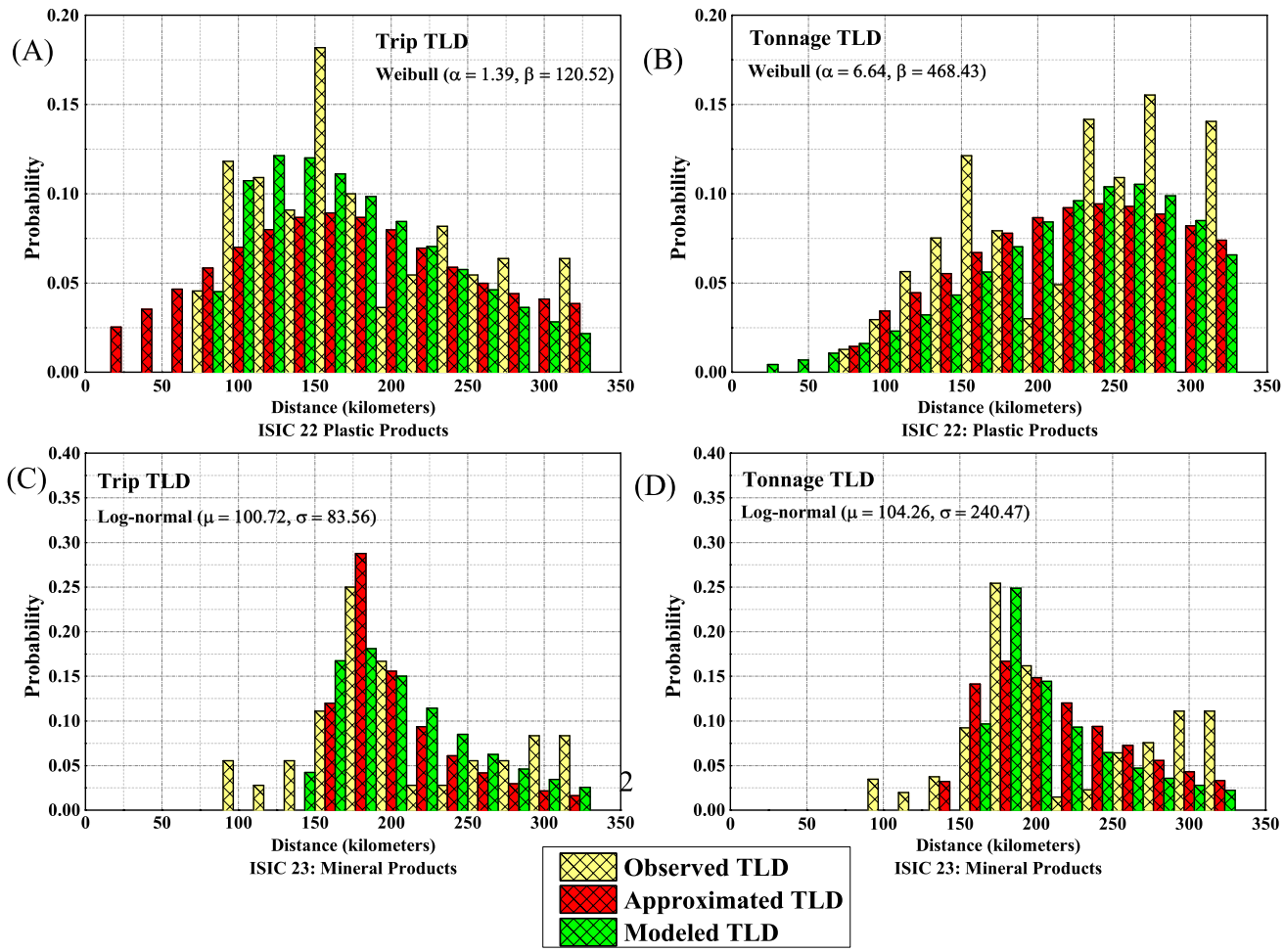

Fig. 5 Trip and tonnage TLD for plastic (**a**, **b**) and mineral products (**c**, **d**)

Log-normal, Gamma, Weibull, Normal, and Exponential. The TLD functions are compared based on RMSE values to suggest the best fit function with the lowest RMSE. Class-specific 'approximation function' such as cubic, quadratic, and log transformation is developed for the trip-tonnage TLD conversion. Lognormal and Weibull PDFs are found to be the best fit for most of the commodity classes in the case of trip TLDs. Tonnage TLDs follow Log-normal, Gamma, and Weibull PDF. Analysis of TLD characteristics across the commodity classes suggests that trip TLD differs from tonnage TLD in each class, as the spatial distribution of trip and tonnage varies for each class. It is also found out that the presence of a port in a city affects spatial distribution leading to long-haul intercity trips for commodities such as food, beverages, metal products, etc.

The research findings and implications can be briefly summarized as follows: (a) there exists no predominant distribution function that can represent the trip and tonnage TLD across all the commodity classes, (b) the trip and tonnage approximation function for the majority of the commodity classes follow the Log transformation equation form, (c) the approximated trip and tonnage TLDs matches to the observed TLDs, which confirms the statistical validity of introducing the concept of 'approximation function' (d) it is possible to cluster certain commodities into a single class based on the similarity in their trip- and tonnage TLDs, (e) the approximation function appears to be a simple approach for trip-tonnage conversion and vice-versa, (f) the concept of approximation function can be considered as a surrogate measure for the tonnage-trip conversion factor known as truck equivalency factor, (g) the approximation function is a crucial input for integrating commodity-based- and trip-based modeling approaches, (h) the modeling integration framework will be useful to urban planners for reducing data reporting burden and saving in data collection resources.

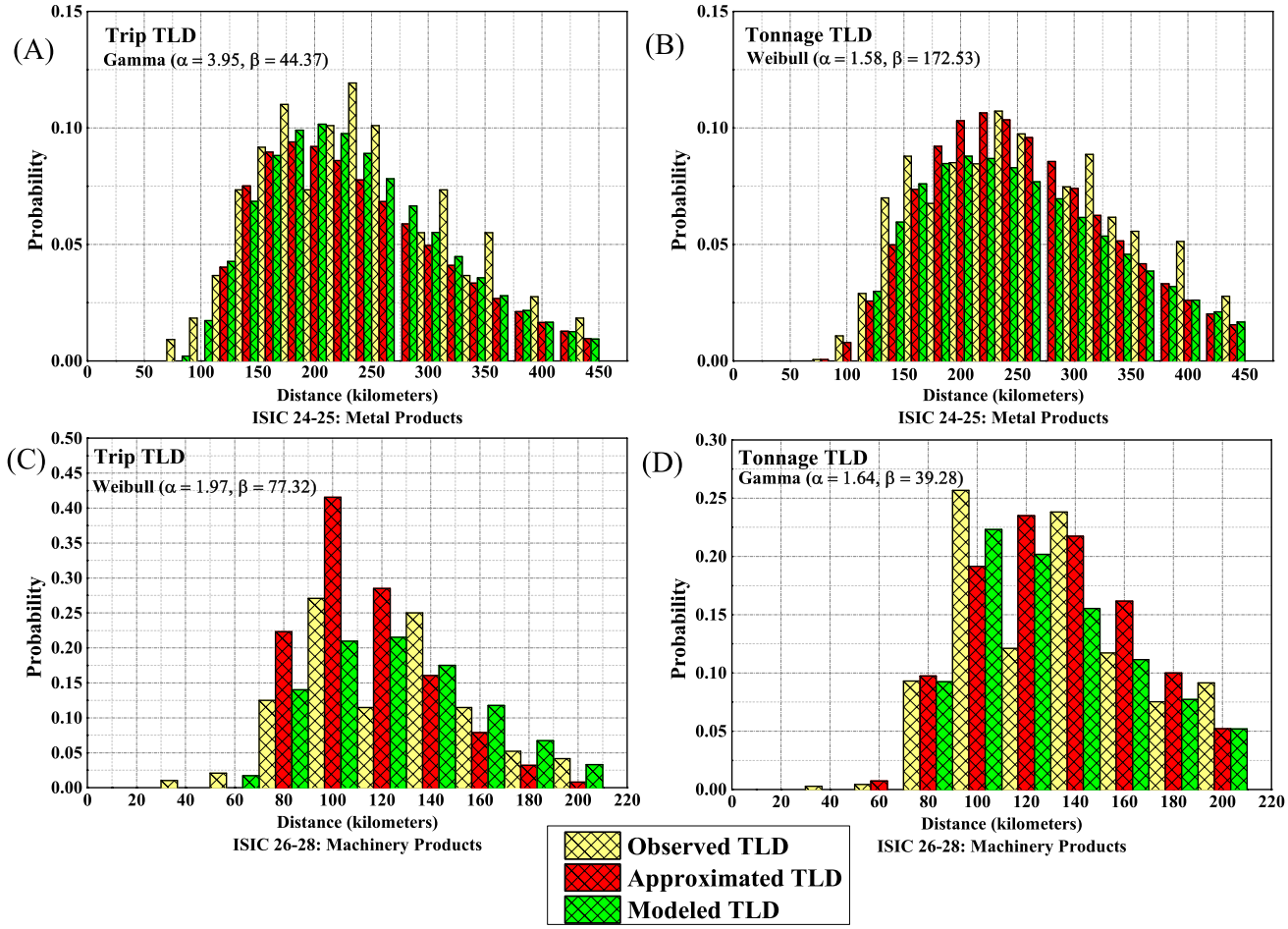

Fig. 6 Trip and tonnage TLD for metal products (**a**, **b**) and machinery products (**c**, **d**)

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Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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